Example:

$S \rightarrow E$

$E \rightarrow E+E \mid E-E$

$E \rightarrow I$

$I \rightarrow id$

string: id-id+id

Right most derivation:

$S \xrightarrow{rm} E$

$\xrightarrow{rm} E+E$

$\xrightarrow{rm} E+I$

$\xrightarrow{rm} E+id$

$\xrightarrow{rm} E-E+id$

$\xrightarrow{rm} E-I+id$

$\xrightarrow{rm} E-id+id$

$\xrightarrow{rm} id-id+id$

Parse Trees:

A tree representation for derivation is called parse tree.

$S \rightarrow E$

$E \rightarrow E+E \mid E-E$

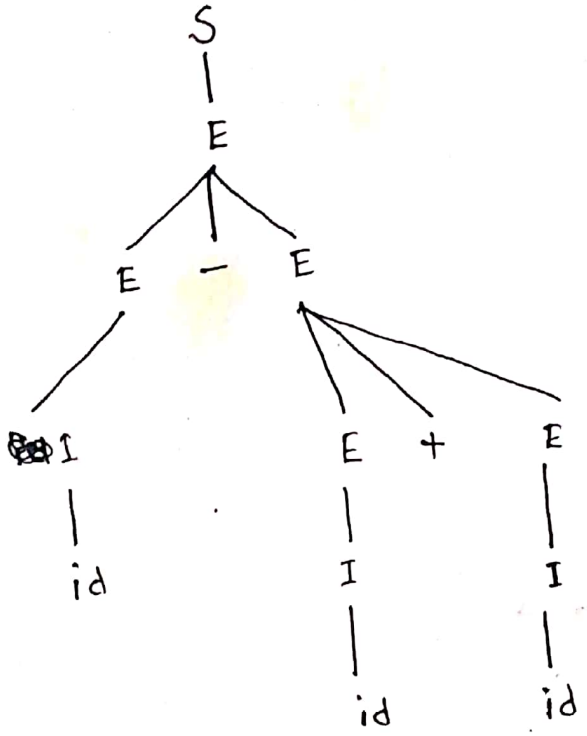
$E \rightarrow I$

$I \rightarrow id$

string: id-id+id

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parse tree:



If we look at the leaves of any parse tree and concatenate them from the left, we get a string, called the yield of the tree.

Example: id - id + id is a yield of the tree

Grammar for regular expression

$0^*1(0+1)^*$

Grammar:

$S \longrightarrow A1B$

$A \longrightarrow \epsilon \mid 0A$

$B \longrightarrow \epsilon \mid 0B \mid 1B$

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